until American ecologist Gene Likens carried out in-depth studies in an area of rural New Hampshire that the phenomenon came to be properly understood.

From 1963 onward, freshwater ecologist Likens and his team studied the relationship between water quality and life forms in the Hubbard Brook drainage basin in New Hampshire. They discovered that the rainfall there was unusually acidic. Acidity, as expressed by pH (potential of hydrogen), ranges from 0 (most acidic) through 7 (neutral), to 14 (least acidic). Most fish and other aquatic animals fare best in water with pH values of 6–8, but Likens found values of 4—too acid for fish, frogs, and the insects they eat to survive. He set up monitoring stations around New England, which showed that acid rain and snowfall were widespread in the densely populated and heavily industrialized northeastern states. Likens's systematic work persuaded the US government to introduce laws to control emissions of the chemicals responsible for acid rain.